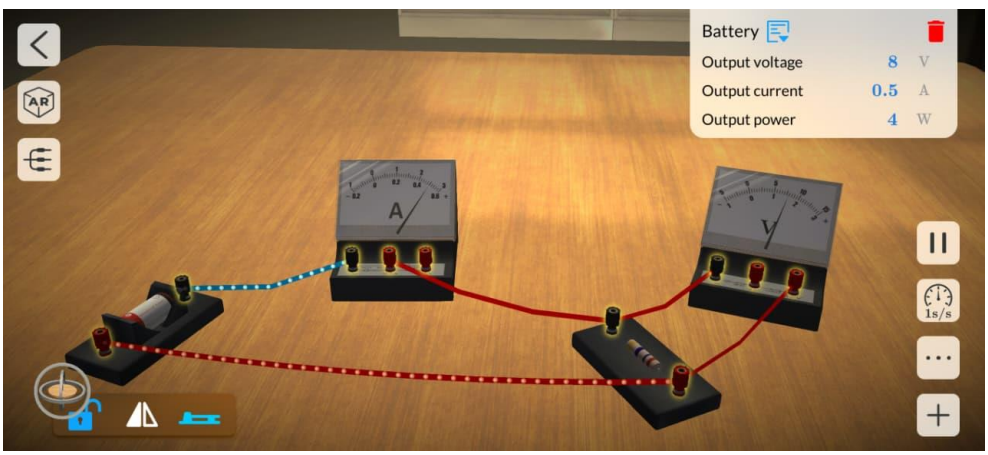
Assessment task: Ezra's Electrical Circuits

Ezra was responsible for checking the electrical safety in the lab and had to test several circuits. In the lab, Ezra discovered four wires connected to unknown resistors. To confirm their identities, Ezra noted the voltage across and the current passing through each resistor and recorded the data. Table 1 shows Ezra's measurements.

Table 1. Data of four resistors in different circuits.

Sample	Voltage (V)	Current (A)	Resistance (Ω)
1	8.0	0.50	16.0
2	9.0	0.75	12.0
3	4.8	0.3	16.0
4	4.8	0.40	12.0



A) Which information in the table would you use to tell Ezra whether any resistors could be the same component? Be sure to tell why.

B) Based on the information in the table, which (if any) of the resistors are the same? Support your answer using your knowledge about relation of voltage, current, and resistance of the circuits and Ohm's Law statement.

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A) Which information in the table would you use to tell Ezra whether any resistors could be the same component?

To determine whether any resistors are the same component, Ezra should look at the resistance (Ω) column in the table.

According to Ohm's Law, $V = I \times R$, it shows that for any given resistor, if we divide the voltage by the current, the result should always be the same number (in this case: resistance), no matter how much voltage or current applied. The resistance is constant for a given component if the ratio of voltage (V) to current (I) remains the same. Therefore, by comparing the resistance values, Ezra can identify which resistors have the same resistance and could have the same component

B) Based on the information in the table, which (if any) of the resistors are the same?

In Ohm's Law: $V = I \times R$, the law says that the voltage across a resistor equals the current mutiplied by its resistance.

If we compare the data:

- Samples 1 and 3 both have 16 Ω resistance, which shows

that they behave the same way when the current passes through. So, they are likely the same resistor

- Samples 2 and 4 both have 12 Ω resistance, so these 2 are also most likely the same resistor no matter the voltage (V) or current (A).

This means that there are 2 pairs of identical resistors:

- Sample 1 and 3 (16 Ω)
- Sample 2 and 4 (12 Ω)